

half image

Description

For some reason a flag image was cut in half Can you find the reason and get the flag?

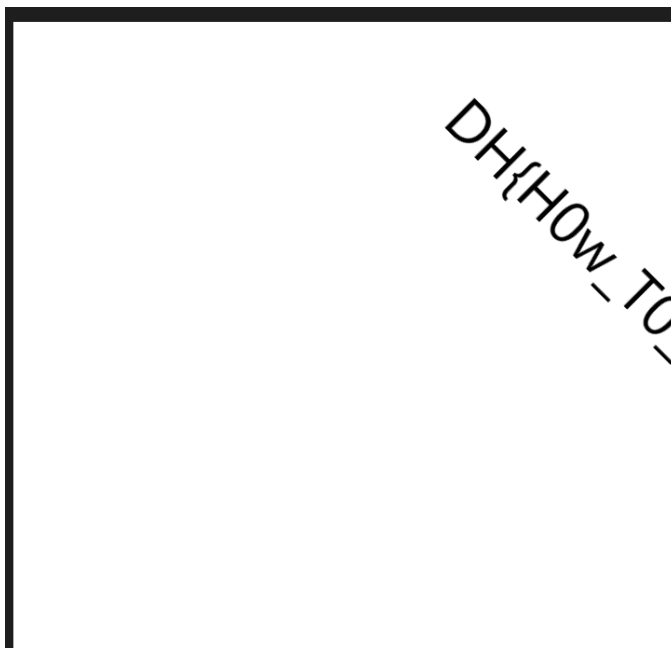
사용한 도구

VS Code

Background

1. 주어진 파일 분석

flag.py → half image



flag.py

```

from PIL import Image
img = Image.open("testfile.png").convert("RGB")
width, height = img.size
hw = width // 2
left = img.crop((0, 0, hw, height))
right = img.crop((hw, 0, width, height))
left.save("left.png")
rdata = right.tobytes()
with open("left.png", "rb") as fl: lp = fl.read()
with open("flag.png", "wb") as f_out:
    f_out.write(lp)
    f_out.write(rdata)

```

2. 익스플로잇 코드 작성

```

from PIL import Image

flag_path = "flag.png"

# 1) 왼쪽 절반: 정상 PNG
left_img = Image.open(flag_path).convert("RGB")
w_left, h = left_img.size

# 2) 파일에서 IEND 이후 숨겨진 데이터 추출
with open(flag_path, "rb") as f:
    data = f.read()

sig = b"\x00\x00\x00\x00IEND\xAE\x42\x60\x82"
iend_pos = data.rfind(sig)
trail = data[iend_pos + len(sig):]

# 3) 오른쪽 이미지 크기 계산
w_right = len(trail) // (h * 3)
right_img = Image.frombytes("RGB", (w_right, h), trail)

# 4) 합치기

```

```
out = Image.new("RGB", (w_left + w_right, h))
out.paste(left_img, (0, 0))
out.paste(right_img, (w_left, 0))
out.save("recovered.png")

print("복원 완료! -> recovered.png")
```

Flag

DH{H0w_T0_Get_Flags?!}